KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION

FitFlow AI – Personalized Yoga & Fitness Companion: An Online Retrieval-Augmented Generation (RAG) & Machine Learning Wellness System

Project ID: PRJ089

Team ID: T089

Team Name: CODE KNIGHTS

Team GitHub Repository: github.com/dhudekulazaheer-jpg/FitFlow-AI

Live Cloud Deployment: https://fitflow-ai-3wf4.onrender.com

Student Name	Registration Number	Individual GitHub Link
Dhudekula Zaheer	99230040524	github.com/dhudekulazaheer-jpg
Jampani Rajesh Babu	99230040574	github.com/rajesh4275
Dasari Bhanu Chaitanya	99230041070	github.com/bhanuchaitanyadasari
Kalukuri Ghandeev Raj	99230041228	github.com/99230041228-art
Kilari Rishi Kumar	99230040337	github.com/kilaririshitharak-svg/Gen-Ai-Assignments

Project Trainer / Guide

Mr. SHIRWIN PRINCE I

Table of Contents

1. Introduction	Section 1
2. Problem Statement & Objectives	Section 2
3. System Architecture (RAG + Safety Pipeline)	Section 3
4. Implementation Details & Tech Stack	Section 4
5. Results and Testing	Section 5
6. Conclusion & Future Scope	Section 6

1. Introduction

Large Language Models (LLMs) have transformed personalized digital assistance, yet unconstrained LLMs frequently suffer from hallucinations and can recommend physically unsafe exercises when used directly for health and fitness guidance. Retrieval-Augmented Generation (RAG) mitigates these risks by grounding model responses in verified domain records prior to generation. FitFlow AI combines an Online RAG + In-Context Learning (ICL) Wellness Coach ('Yogi Coach') with supervised Machine Learning calorie regression and deterministic clinical safety guardrails to generate personalized, injury-aware yoga and workout plans.

2. Problem Statement & Objectives

Problem: Conventional fitness applications rely on static, one-size-fits-all templates that overlook individual joint injuries, surgical recovery histories, and available equipment. Conversely, ungrounded AI chatbots risk hallucinating strenuous poses or movements that can aggravate knee, back, shoulder, or neck conditions.

Objectives:

- Build an Online RAG + ICL coaching engine ('Yogi Coach') strictly grounded in curated yoga, workout, and clinical rule datasets.
- Implement vector semantic retrieval using TF-IDF (bigrams) and Cosine Similarity with automated injury contraindication filtering.
- Enforce deterministic clinical safety gates (pausing generation for uncleared surgeries) and inline source citations ([Doc X: Name]).
- Integrate supervised Random Forest calorie estimation, 7-day structured splits, progressive overload tracking, and cloud database persistence.

3. System Architecture (RAG + Safety Pipeline)

The architecture of FitFlow AI's RAG subsystem operates across two primary phases: Offline Knowledge Indexing and Real-Time Query, Clinical Safety Gating, and Online LLM Generation.

3.1 Data Ingestion & Vector Indexing Phase

On server startup, the RAG retriever (*ml/rag_retriever.py*) ingests and normalizes **1,373 curated records** across three structured knowledge sources: **48 yoga poses** (*datasets/yoga/yoga.json*), **1,318 unique workout exercises** (*datasets/workout/exercises.json*), and **7 clinical joint precaution guidelines** (*ml/health_rules.json*). Each record is converted into a rich textual document combining anatomy targets, equipment, benefits, and step-by-step instructions, then vectorized via Scikit-Learn's TF-IDF Vectorizer (1–2 n-grams, English stop-word removal, sublinear TF scaling).

3.2 Query, Clinical Safety Gate & Generation Phase

When a user submits a request to Yogi Coach, the backend first runs the deterministic clinical safety evaluator (*ml/health_safety.py*) against the user's saved health assessment in Supabase PostgreSQL. If cleared, the vector retriever computes Cosine Similarity between the query and indexed corpus, excludes contraindicated movements (e.g., plyometrics for knee pain), and injects the top-K retrieved documents alongside user biometrics into an In-Context Learning (ICL) prompt sent to the online LLM (Groq / Google Gemini / OpenAI) for grounded generation.

4. Implementation Details & Tech Stack

Component	Technology Used	Purpose
Backend & Security	Python 3.10+, Flask, Gunicorn, Flask-WTF	REST APIs, CSRF protection, authentication, and session isolation
Vector Store & Retrieval	Scikit-Learn TF-IDF + Cosine Similarity	Semantic search over 1,373 yoga, workout, and clinical rule records
RAG + ICL Orchestration	Custom Pipeline (ml/rag_engine.py)	Safety gating, biometric prompt augmentation, and source citations
Large Language Model	Groq API (openai/gpt-oss-120b) / Gemini	Online natural-language routine synthesis strictly from context
Predictive ML Model	RandomForestRegressor (300 trees)	Calorie burn prediction trained on Kaggle gym exercise dataset
Cloud Database & Hosting	Supabase PostgreSQL (psycopg3) & Render	Persistent SSL cloud storage and public HTTPS web deployment

4.1 Sample Vector Indexing & Retrieval Configuration

To enable fast semantic lookup alongside clinical safety filtering without exceeding cloud memory constraints, the retriever vectorizes all curated records with n-gram TF-IDF and cosine similarity:

```
vectorizer = TfidfVectorizer(ngram_range=(1, 2), stop_words="english",
sublinear_tf=True)
tfidf_matrix = vectorizer.fit_transform([doc["text"] for doc in documents])
similarities = cosine_similarity(vectorizer.transform([enriched_query]),
tfidf_matrix)
```

5. Results and Testing

FitFlow AI was verified end-to-end across both automated test suites and live cloud deployment on Render (*https://fitflow-ai-3wf4.onrender.com*) connected over SSL to Supabase PostgreSQL:

- **Automated Verification Suite (36/36 Passing):** 16 baseline handover & isolation tests (*test_handover.py*), 5 recovery & exercise swap tests (*test_new_features.py*), 9 weekly program & social leaderboard tests (*test_advanced_features.py*), and 6 Online RAG + ICL & Floating Yogi Coach UI tests (*test_rag_feature.py*).
- **RAG Knowledge Coverage & Grounding:** Indexed 1,373 curated records (48 yoga poses, 1,318 exercises, 7 safety rules). Grounding LLM prompts strictly in retrieved dataset chunks with inline [Doc X] citations eliminates unverified external exercise suggestions and enforces injury filtering.
- **Calorie Prediction Model Performance:** Our Random Forest Regressor achieved **R^2 = 0.8602** (86.02% variance explained), **MAE = 85.16 kcal**, and **RMSE = 108.00 kcal** on held-out Kaggle test data—reducing MAE by **63.16%** compared to the Mean Baseline (MAE = 231.20 kcal).
- **Interactive & Social Features Verified:** Transparent two-step Yogi Coach UI with pure HTML/CSS animated floating avatar, Web Audio interval coaching chimes, and privacy-preserving friend streak leaderboards.

6. Conclusion & Future Scope

Conclusion: FitFlow AI successfully bridges classical predictive machine learning, clinical health safety logic, and modern Generative AI via Retrieval-Augmented Generation (RAG) and In-Context Learning (ICL). By grounding online LLM responses in 1,373 curated exercise and yoga records and persisting user biometrics securely in cloud PostgreSQL, the system delivers safe, transparent, and actionable fitness coaching.

Future Scope: Planned enhancements include hybrid dense vector embeddings (e.g., pgvector inside Supabase) for deeper multi-lingual semantic recall, real-time smartwatch/wearable heart-rate integration for adaptive calorie and recovery tracking, camera-based computer-vision posture correction during live workouts, and automated retrieval-quality evaluation via user session feedback loops.